



SIMATS ENGINEERING

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Course Code: DSA0405

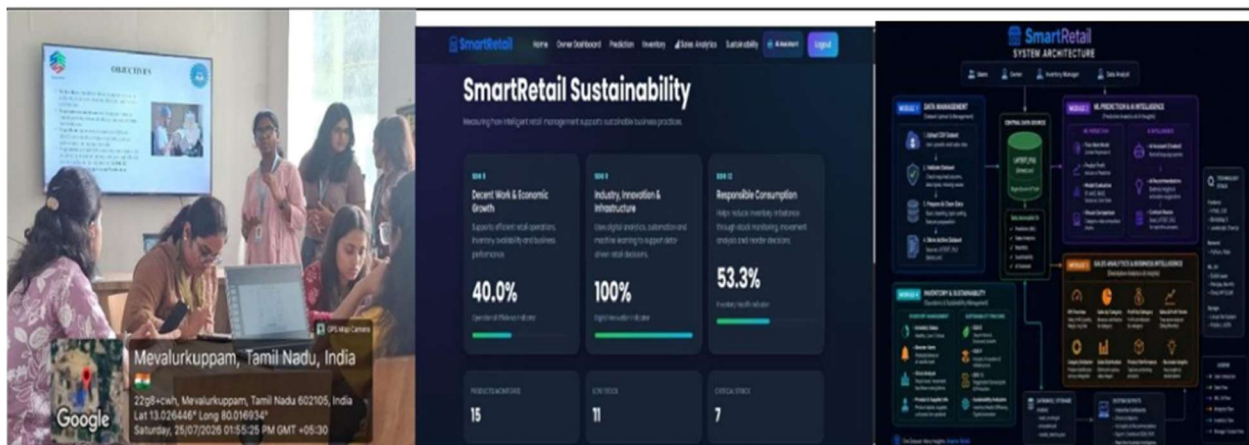
Slot: D

Course Name: Fundamentals of Data Science

Course Faculty: Dr KrishnaRaj Mony

Project Title: Smart Retail: A Data Driven Decision Support System for Sustainable Retail Operations

Module Photographs:



Project Description:

SmartRetail is a data-driven retail decision support system designed to help businesses analyze sales, predict profits, manage inventory, and improve decision-making. The system integrates data preprocessing, exploratory data analysis, machine learning, business intelligence, and AI-based recommendations into a single platform. It provides interactive dashboards for monitoring sales trends, product performance, profit, and inventory status.

SmartRetail is an integrated retail analytics platform developed to simplify and improve retail business management through data-driven techniques. The system processes retail datasets, cleans and prepares the data, and generates meaningful insights through interactive visualizations. It uses machine learning to predict profit and evaluate model performance using different metrics. The platform also provides sales and product analysis, helping businesses identify profitable categories, sales trends, and top-performing products. An AI-powered assistant enables users to interact with the system using natural language and receive useful business insights. In addition, inventory monitoring and sustainability tracking help reduce stock-related issues and encourage responsible resource management.

Student Signature

Guide Signature